

ECON567B: Econometrics II

Southern Illinois University Carbondale

Fall 2026

INSTRUCTOR	Soo Jeong Lee School of Analytics, Finance and Economics Southern Illinois University Office: Faner Hall Room 4136 Carbondale, IL 62901 Email: soojeong.lee@siu.edu Website: soo-econ.github.io Student Engagement Hours: Tue. 1-1:50 PM or by appointment
TEACHING ASSISTANT	▪ Teaching Assistant/Email: TBA ▪ Student Engagement Hours/Location: TBA
LECTURE	08/17/2026 - 12/11/2026 ▪ Tue. Thu. 2:00 PM - 3:15 PM ▪ Faner Hall Room 3512 ▪ No class: November 24 and 26 (Thanksgiving Break)
COURSE GOAL	▪ The second semester of Ph.D.- level sequence in Econometrics ▪ Prerequisite: ECON 465 and ECON 567A, or consent of instructor. The aim of this course is to develop students' understanding of econometric tools most often used in applied research, with a focus on model specification and data issues. While theoretical foundations will be covered as needed, the primary emphasis is on practical application—understanding the assumptions behind each method, interpreting results, and implementing techniques using real-world dataset. ▪ Course Sequence: ECON 567C To help students see the full scope of the sequence and plan their studies, the course outline provided in the following pages includes all topics that I aim to cover across both ECON 567B and ECON 567C.
STUDENT LEARNING OUTCOMES	By the end of this course, students will be able to: ▪ Explain the key identifying assumptions behind econometric methods ▪ Choose appropriate econometric models based on data and research questions. ▪ Interpret results from econometric analyses in a clear and meaningful way. ▪ Develop a research proposal that includes a clear question, model specification, and identification strategy.
GRADING	The course grade will be based on the following components: ▪ In-class quizzes and assignments: 60% (tentative: 3 quizzes and 3 assignments) ▪ Referee report: 20% ▪ Presentation/ Research Proposal: 30% Students will give one presentation and complete two essays: a referee report and a research proposal. Statistical computing is required for the assignments. Students may use any programming language of their choice. Please note that course examples will primarily use Stata.

**COURSE
MATERIALS**

This course is primarily based on a **set of lecture notes**. Students are required to read the assigned sections from the following main textbook.

- **Main Textbook:** Wooldridge, J.M. (2010), *Econometric Analysis of Cross Section and Panel Data*, 2nd edition, MIT Press, Cambridge, MA.

**COURSE
OUTLINE**

Course Topics (EC567B, EC567C)

ECON567B, Econometrics II

1. Math Review Wooldridge: Ch. 1-3
2. OLS Estimation with Cross Sectional Data Wooldridge: Ch. 4
3. Systems of Equation: Pooled OLS (Panel Data) Wooldridge: Ch. 7
4. Systems of Equation: Generalized Lease Squares (GLS) Wooldridge: Ch. 7
5. Systems of Equation: Instrumental Variables (IVs) Wooldridge: Ch. 8
6. Linear Unobserved Effects Panel Data Models Wooldridge: Ch. 10-11
: Random Effects vs. Fixed Effects Estimation
7. Treatment Effect Estimation under Unconfounded Assignment Wooldridge: Ch. 21

ECON567C, Econometrics III

1. Difference in Differences with Panel Wooldridge: Ch. 21
: Common Timing, Staggered Intervention
 2. Regression Discontinuity Designs Wooldridge: Ch. 21
 3. Maximum Likelihood Estimation Wooldridge: Ch. 13
 4. Binary Response Models: Cross Sectional and Panel Data Wooldridge: Ch. 15-17
 5. Exponential Models for Nonnegative Response Variables Wooldridge: Ch. 18
 6. Treatment Effect Estimation under Confounded Assignment Wooldridge: Ch. 6, 21
: Local Average Treatment Effects (LATE), IV and Control Function Approaches
 7. Time Series Analysis
 8. Introduction to Machine Learning
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